How array geometry shapes the orientation sensitivity of glider-derived vertical velocity

Technical note — TPOSE24 wave-glider OSSE

August 12, 2026

Abstract

A small array of wave gliders estimates horizontal divergence—and, through continuity, vertical velocity w—by fitting a plane to the velocities it samples and reading off the slopes. Because the ocean features drifting through the array arrive at arbitrary orientations, we ask a design question: *does the estimate depend on how the flow happens to be oriented relative to the array, and if so, which array shapes are least sensitive?* We work up from specific cases to the general one. Single-scale test flows turn out to fool exactly one array shape each and leave every other shape exactly invariant; a broadband, ocean-like flow fools every shape a little, in a ranking that at first looks erratic—a pentagon beating a hexagon, a triangle tying it. Averaged over where the array sits in the flow, however, the ranking becomes strictly monotone in the number of gliders. All of it follows from one idea: n gliders equally spaced around a ring sample the flow at n azimuthal points, and like any equally spaced sampling they cannot resolve structure that repeats a multiple of n times around the ring. The number of gliders therefore sets which azimuthal scales alias, exactly as a sampling rate sets a Nyquist frequency. Together with noise, robustness, and footprint considerations, this favors the regular hexagon for a six-glider array.

1 The estimate and the question

Let the gliders sit at horizontal offsets (x, y) (east, north) from the array center, all at radius r from it, so the array diameter is $d = 2r$. The array samples the horizontal velocity (U, V) and estimates the horizontal divergence

$$D = \partial U/\partial x + \partial V/\partial y, \qquad w(-H) = -\int_{-H}^{0} D \, \mathrm{d}z, \tag{1}$$

the vertical velocity following by integrating continuity down to depth H. With only a handful of point samples, the array fits a plane to each velocity component,

$$U \approx a_0 + a_1 x + a_2 y, \qquad V \approx b_0 + b_1 x + b_2 y, \tag{2}$$

and reads the divergence off the fitted slopes, $\widehat{D} = a_1 + b_2$. A plane has a single, constant slope, so $\widehat{D}$ is the array's best single-number estimate of the divergence averaged over its footprint. Because both the estimate and the average are linear in the velocity, time-averaging passes straight through them and we may reason about one snapshot at a time.

The ocean structures that drift through the array—eddies, fronts, wave packets—arrive at every orientation. So we rotate the flow rigidly by an angle θ relative to the fixed array and ask how $\widehat{D}$ responds. An array whose estimate is independent of θ is one whose skill does not depend on the (unknowable) orientation of the passing feature. Throughout we compare the five regular arrays of Fig. 1, all at the same radius $r = 0.75°$, so that differences come from the *arrangement* of the gliders rather than from the size of the array.

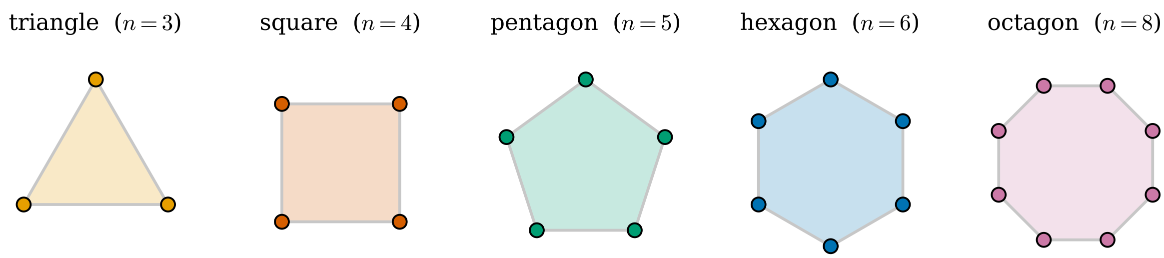

Figure 1: The five regular arrays compared throughout, drawn at a common radius r. Only the number of gliders n and their angular arrangement differ.

2 One flow, two arrays

Start with a single concrete case (Fig. 2): a sum of two tilted waves with structure comparable in scale to the array, sampled by the two candidate OSSE footprints—the four-corner square and the regular hexagon. The flow (panel a) has a curved divergence field across the footprint, with a positive core and negative flanks (panel b), so no plane can reproduce it exactly.

Rotating that flow past the arrays (panel c) shows the effect this note is about. The estimate is not a fixed number: it depends on the orientation at which the feature arrives. The square's error swings by $6.4\times10^{-7}\ \mathrm{s}^{-1}$ peak to peak and cycles *four* times per revolution, with a smaller ripple at eight cycles superposed; the hexagon swings by only $1.6\times10^{-7}\ \mathrm{s}^{-1}$—four times less—and cycles *six* times. Both numbers, four and six, will turn out to be exactly the glider counts, but let us treat them as measurements for now.

Two things to keep separate. Both curves in Fig. 2(c) are offset *above* zero (means +0.9 and $+0.5\times10^{-7}\ \mathrm{s}^{-1}$). That offset is the same at every orientation: it is the error a flat plane makes against a curved field, a scale-matching problem set by array size (§7), and it has nothing to do with orientation. The *swing* about that offset is the orientation sensitivity. From here on we measure the swing as the peak-to-peak range of $\widehat{D}(\theta)$ about its own rotational mean, which isolates the orientation dependence and requires no choice of reference truth; it is also directly comparable between shapes whose footprints enclose different areas. Panel (c) is the one place we grade against the truth (each array's own footprint average), because there the offset itself is worth seeing.

3 Single-scale flows: each fools exactly one shape

Next take flows simple enough to isolate one behavior at a time (Table 1) and run all five arrays through a full rotation (Fig. 3). The results are unusually clean.

- A flow with a **uniform gradient** is reproduced perfectly by every array at every orientation. A plane fit is exact for a field that really is a plane, so the first thing it can get wrong is *curvature*.
- The **curvature** flow moves the *triangle* and nothing else. The triangle's estimate cycles three times per revolution; the square, pentagon, hexagon and octagon are flat to machine precision—exactly invariant.
- The two **cubic** flows move the *square* and nothing else, cycling four times per revolution (period 90°), in opposite phase to each other. The triangle, pentagon, hexagon and octagon are again exactly invariant.

The headline is counter-intuitive: the square—the roundest-looking four-point array—is the *only* regular polygon that fails the cubic flows, and it fails them precisely because its four-fold symmetry matches something four-fold in the flow. Symmetry that matches the flow is a liability, not an asset. Note also the pattern in Table 1: whether the velocity reverses sign across the array center decides whether the flow has an even

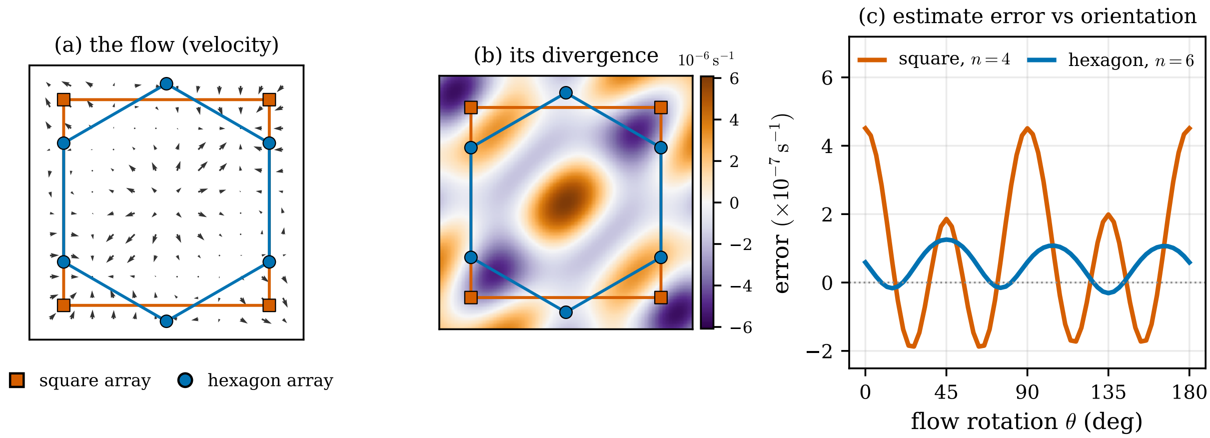

Figure 2: A worked example. *(a)* A two-scale wave flow, with the four-corner square array (1.5° box) and the regular hexagon drawn on top. *(b)* Its true divergence, which is curved across both footprints. *(c)* Error of each array's plane-fit divergence—estimate minus the true average over that array's own footprint—as the flow is rotated through 180°. The square cycles four times and swings four times further than the hexagon, which cycles six times. The common offset above zero is the orientation-independent bias of a plane fit to a curved field.

or an odd number of lobes, and that in turn decides which shape it catches. We come back to this in §6; it is the single most consequential fact in the note.

4 Broadband flow at one placement

Real ocean flow is not single-scale. Returning to the two-scale wave field of §2—still with the array centered exactly as in Fig. 2—no array is exactly invariant any more, and the ranking (blue bars in Fig. 4a) looks erratic:

	triangle	square	pentagon	hexagon	octagon
swing ($\times 10^{-7}\,\mathrm{s}^{-1}$)	3.66	9.81	0.15	3.66	3.02

The square is by far the worst. The pentagon is essentially silent. The triangle—three gliders—ties the six-glider hexagon *exactly*, to five decimal places. This is not what one expects from "more gliders is better,"

Test flow (case)	U (V: swap $x \leftrightarrow y$)	reverses across the center?	error cycles per flow revolution	shape it fools
uniform gradient (linear)	$U = S\,x$	yes	—	none — plane fit is exact
curvature (even)	$U = C\,x^2$	**no**	3	**triangle**
cubic (odd)	$U = C\,x^3$	yes	4	**square**
cubic + cross-term (mixed)	$U = C\,(xy^2 + \frac{1}{2}x^3)$	yes	4	**square**
single wave (fine)	$U = C\,\sin(Kx + By)$	yes	$2, 4, 6, \ldots$	every shape (a little)
two waves (combo)	sum of two such waves	yes	$2, 4, 6, \ldots$	every shape

Table 1: The synthetic test flows, and what is *observed* when each is rotated past the five arrays. "Error cycles per flow revolution" is read directly off Fig. 3. Nothing here is theory yet: the last two columns are measurements.

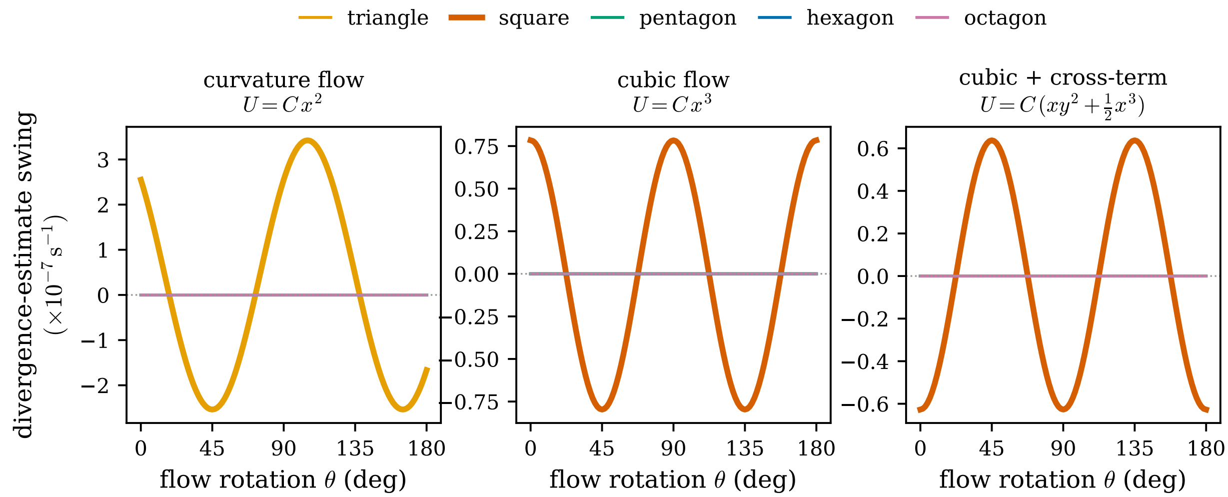

Figure 3: Divergence-estimate swing versus the flow-rotation angle θ for the three polynomial flows, all five arrays overlaid. The triangle alone oscillates for the *curvature* flow; the square alone oscillates for the *cubic* and *cubic + cross-term* flows, in opposite phase; every other array is flat—exactly invariant, to machine precision. Bold curves mark the shape that is fooled in each panel.

and it is worth being suspicious of: the ranking is not monotone in n, and one of these numbers is about to turn out to be an accident.

Adding a centered feature. Real flow also contains features the array straddles whose velocity does not reverse across the center—a jet core, an eddy center. Adding one to the wave field, scaled to the same RMS speed (red bars, Fig. 4a), raises the triangle's swing from 3.66 to $12.3 \times 10^{-7}\ \mathrm{s}^{-1}$ and leaves the square, pentagon, hexagon and octagon *numerically unchanged*. Which shape responds is a property of the geometry; how large the response is depends on how energetic the feature happens to be, and Fig. 4(b) sweeps that: four flat lines and one rising one, with the triangle overtaking the square only past about 0.75 of the wave field's RMS speed. So the honest claim is not "the triangle is the worst shape"—in this flow the square is worse until the feature is strong—but that the triangle is the only array a centered feature can touch at all.

5 The general case: many placements

The suspicious numbers in §4 come from one particular placement: the array sitting exactly on the center of the wave field, straddling a zero crossing, so that the velocity it sees reverses across the array. That is a symmetric special case, and the ocean has no reason to oblige.

Before averaging anything, take one specific alternative. Displace the array by $(0.37°, 0.21°)$—about a third of the dominant wavelength, so that it no longer straddles the zero crossing—and run the identical rotation experiment in the identical flow:

	triangle	square	pentagon	hexagon	octagon
centered, from §4	3.66	9.81	0.15	3.66	3.02
displaced ($\times 10^{-7}\ \mathrm{s}^{-1}$)	4.16	7.84	6.14	2.10	1.36

The pentagon goes from the quietest shape to the second loudest—0.15 to 6.14, a factor of forty—and now loses to the hexagon three times over. The triangle's exact tie with the hexagon is gone as well (4.16 against

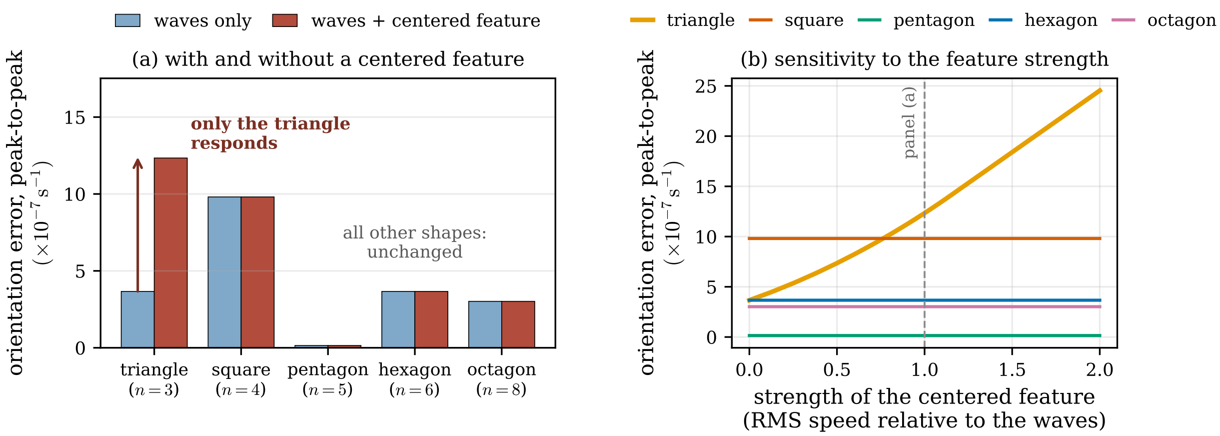

Figure 4: Orientation error (peak-to-peak swing over a full flow rotation) for the broadband, two-scale wave field at the centered placement, with and without a centered feature (a jet/eddy core, whose velocity does not reverse across the array). *(a)* Adding the feature raises the error of *only* the triangle; the other four are unchanged to machine precision. Note that the square is nonetheless the loudest shape on the waves alone, and that the ranking is not monotone in n. *(b)* The same error as the feature's strength is swept: four shapes are flat—geometrically insensitive to it—while the triangle rises and overtakes the square at about 0.75. Which shape responds is set by geometry; which is worst overall is set by the flow.

2.10). Nothing about the arrays changed between these two rows: same shapes, same radius, same flow, same rotation. Only where the array sits in the field changed. Whatever produced the striking ranking of §4, it was not a property of the shapes.

Figure 5 does this systematically, for 400 random placements of each array in the same wave field. Averaged over placement, the ranking is strictly monotone in the number of gliders:

	triangle (n = 3)	square (4)	pentagon (5)	hexagon (6)	octagon (8)
mean swing ($\times 10^{-7}\,s^{-1}$)	8.64	6.12	4.04	3.09	1.60

Two of the puzzles from §4 dissolve. The pentagon's near-silence (0.15) was a property of that one placement, not of the shape: over the ensemble it averages 4.04 and loses to the hexagon, as the glider count says it should. And the triangle's tie with the hexagon was likewise placement-specific—in general the triangle is the *worst* of the five, at more than twice the hexagon's error. The stars in Fig. 5 mark the special placement: lucky for the triangle and pentagon, unlucky for the square and octagon, and representative for none of them.

This is the practically important statement. If an array could be guaranteed to sit centered on the features it measures, shape would matter in the intricate way §4 suggests. It cannot, so what matters is the ensemble: *more gliders, monotonically better*, with the triangle worst.

6 Why: the array as an azimuthal sampler

One idea accounts for every result above. Picture the flow's structure as seen going once around the ring of gliders. It has some azimuthal pattern, which we decompose into modes that repeat m times per lap—m is the azimuthal wavenumber, familiar from wavenumber-1 and wavenumber-2 asymmetries in a vortex; $m = 0$ is uniform.

Equally spaced samples alias. Because n gliders sit at equal angular spacing, summing any azimuthal mode around the ring cancels it *unless* the mode repeats a whole number of times between neighboring

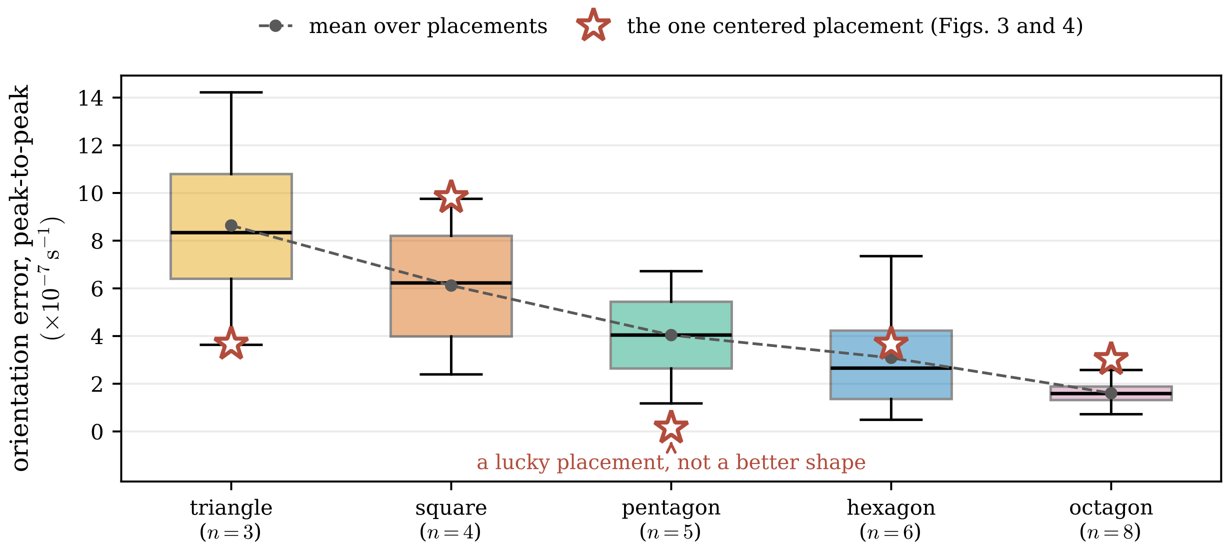

Figure 5: Orientation error over 400 random placements of each array within the same two-scale wave field (boxes: median and interquartile range; whiskers: 5th–95th percentiles). Averaged over placement the ranking is monotone in the number of gliders, and the pentagon's apparent advantage in Fig. 4 disappears. Stars mark the single centered placement used in Figs. 3 and 4, which is an outlier for four of the five shapes.

gliders—that is, unless m is a multiple of n. This is the rotational form of the Nyquist statement: equally spaced samples cannot distinguish structure at multiples of the sampling rate from a constant, and alias it. A regular n-glider array therefore *reproduces the true divergence for any flow with no structure at* $m = n, 2n, 3n, \ldots$, and is fooled only at those *blind modes*. Figure 6 shows the mechanism directly: when the flow's lobes line up with the gliders, every glider reads the same phase, the pattern is indistinguishable from a uniform one, and it leaks into the divergence estimate.

As the flow rotates, the aliased modes sweep past the gliders and the estimate oscillates:

$$\widehat{D}(\theta) = D_{\text{true}} + \sum_{j\geq 1} a_{jn} \cos\big(jn\,\theta - \varphi_{jn}\big) \;\approx\; D_{\text{true}} + a_n \cos\big(n\theta - \varphi_n\big), \tag{3}$$

where a_m is the strength of the flow's structure at wavenumber m. The sum runs only over multiples of n, and because a smooth flow's a_m fall off with m, the lowest surviving term dominates. This is why the square's error in Fig. 2 cycles four times per revolution and the hexagon's six: the cycle count *is* the glider count.

A regular n-glider array's orientation error is set by how much of the flow's structure repeats exactly n times around the ring. More gliders push that blind mode to higher m, where a smooth flow carries less energy—the rotational form of anti-aliasing.

Wavenumber and physical scale. Azimuthal wavenumber ties to physical scale through the array size: a feature of horizontal wavelength λ wrapped around a ring of diameter d registers at

$$m \approx \frac{2\pi r}{\lambda} = \frac{\pi d}{\lambda}. \tag{4}$$

Features much larger than the array ($\lambda \gg d$) sit at low wavenumber and are handled cleanly; features comparable to or smaller than the array reach $m \gtrsim 3$ and begin to alias on the smallest arrays. Two design levers follow: adding gliders raises the blind mode (Eq. 3), and shrinking the array raises the wavelength at which a given feature aliases (Eq. 4), at the cost of a shorter baseline and noisier slopes (§7).

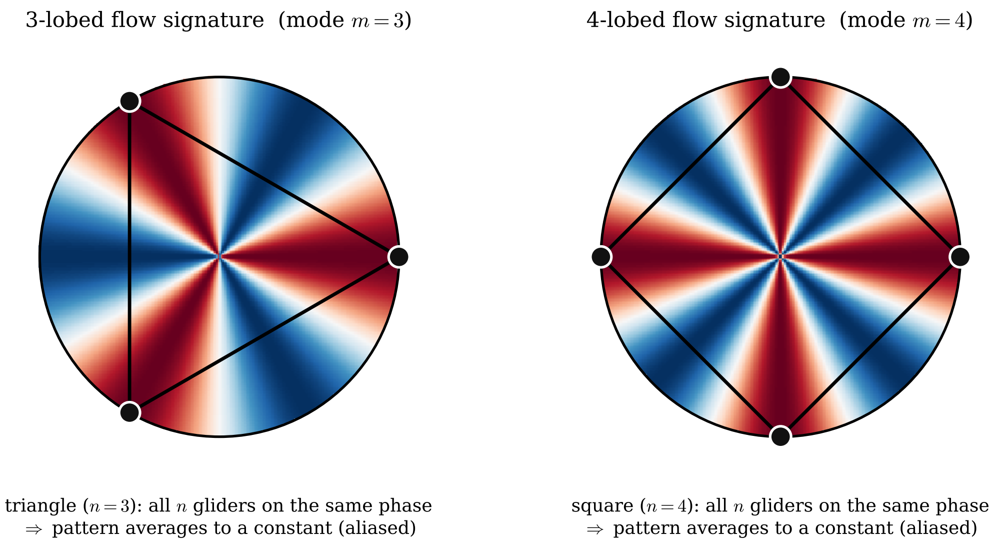

Figure 6: Why a matched flow aliases. *Left:* a three-lobed flow signature ($m = 3$) sampled by a triangle—all three gliders sit at the same phase (here, all on a red lobe), so the three-lobed pattern is indistinguishable from a uniform one and contaminates the divergence estimate. *Right:* the same for a four-lobed signature ($m = 4$) sampled by a square. Each regular array is blind precisely to the wavenumber that matches its glider count.

Parity, and why the pentagon looked good. Here is the fact flagged in §3. If the velocity *reverses* across the array center, the flow's signature contains only even wavenumbers, $m = 2, 4, 6, \ldots$; if it does *not* reverse—a centered jet or eddy core—it contributes odd ones, $m = 1, 3, 5, \ldots$. Now combine that with the blind-mode rule. For a flow that reverses, an array with an *odd* number of gliders is blind at $m = n$ but the flow has nothing there, so its lowest *active* blind mode is $2n$; an array with an *even* number is exposed at n itself. Table 2 works this out, and it explains §4 exactly: at the centered placement the error is ordered by lowest active mode, not by n. The pentagon is quietest because it is not exposed until $m = 10$; the triangle ties the hexagon because both are first exposed at $m = 6$; the square is worst because it is exposed at $m = 4$.

Figure 7 shows this happening, rather than merely asserting it. Panel (a) is the azimuthal spectrum of what the array actually samples—the scalar $xU + yV$ around the glider ring, whose wavenumber-m amplitude is what Eq. (3) turns into an orientation swing, so the vertical axis is plotted directly in swing units and each array's error can be read off at $m = n$. Centered on the field (blue), the spectrum is *exactly* zero at every odd wavenumber: there is nothing at $m = 3$ for the triangle to alias and nothing at $m = 5$ for the pentagon, so both are silent at their own blind modes and only respond at $m = 6$ and $m = 10$ respectively. That is the whole explanation of §4, including the exact triangle–hexagon tie: both are driven by the same $m = 6$ amplitude. Displace the array (red) and the odd wavenumbers fill in—$m = 1$ becomes the single largest component—and the pentagon inherits a substantial $m = 5$. Panel (b) confirms that this accounts for the measurements: the swing predicted from the lowest blind mode carrying any signal (black dashes) reproduces the measured error shape by shape, at both placements.

So, to answer the question directly: *the pentagon does not beat the hexagon in general.* It beats it only for flow that is antisymmetric about the array center, a symmetry no real deployment can count on, and the advantage vanishes as soon as the array is displaced by a fraction of a wavelength. The same caveat cuts the other way for the triangle: its tie with the hexagon is the same accident, and a centered feature—the very thing that breaks the symmetry—exposes it at $m = 3$, the lowest blind mode of any shape considered here.

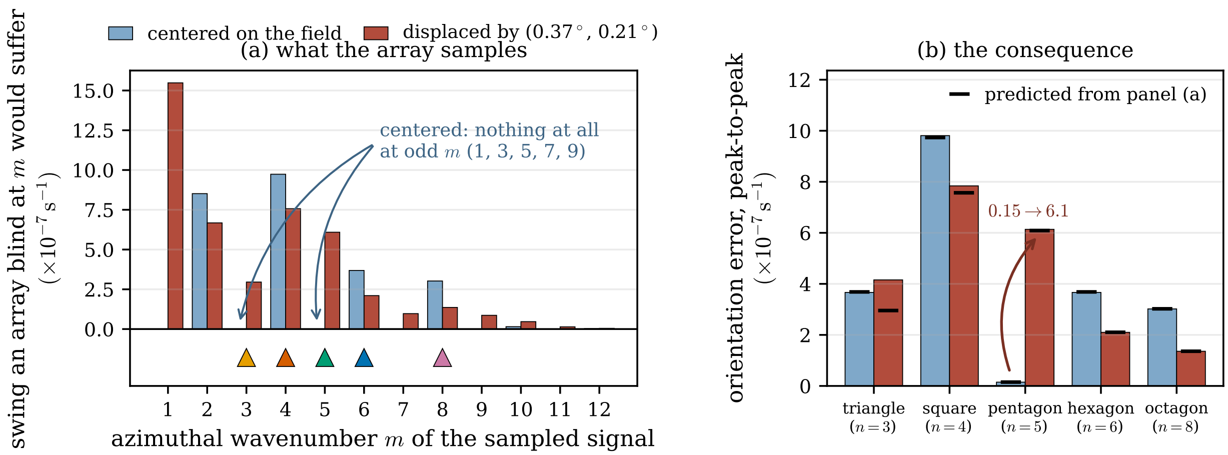

Figure 7: Why parity decides the ranking, and what changes when the array moves. *(a)* Azimuthal spectrum of the signal the array samples ($xU + yV$ around the ring), plotted in swing units so that an array whose lowest active blind mode is m suffers the plotted error; triangles below the axis mark each array's first blind mode $m = n$, colored as in Fig. 1. Centered on the wave field (blue) there is *no* odd-m content at all, so the triangle ($m = 3$) and pentagon ($m = 5$) are blind at wavenumbers the flow does not carry; displaced by $(0.37°, 0.21°)$ (red), odd wavenumbers appear. *(b)* The consequence shape by shape, with the swing predicted from panel (a) marked in black. The pentagon's error rises by a factor of forty. The one visible gap—the displaced triangle—is where two blind modes ($m = 3$ and $m = 6$) contribute comparably and the leading term alone undershoots.

7 What else sets the error, and robustness to array shape

The aliasing picture governs the *orientation* sensitivity. Two other contributions do not depend on orientation and should not be conflated with it:

- **The uniform ($m = 0$) bias.** A flat plane cannot equal the footprint average of a *curved* divergence field, independent of orientation or glider count—the offset seen in Fig. 2(c). This is a scale-matching error (array size versus feature size), reduced by shrinking the footprint (Eq. 4) or by fitting curvature rather than a plane, which requires more than three gliders.
- **Noise.** Random per-glider velocity error of variance σ^2 propagates to $\mathrm{Var}(\widehat{D}) \approx 4\sigma^2/(n\,r^2)$: doubling from three gliders to six halves the variance. A triangle is exactly determined (three points fix a plane, no redundancy); a hexagon is over-determined and averages the noise down.

Finally, the invariance results are robust to array *shape*, not just to regular polygons. An axis-aligned stretch of a hexagon—the same six-glider pattern squeezed in latitude—remains exactly invariant to the cubic flow at *any* elongation; the mechanism-test footprint, which uses a square bounding box (and so is a stretched, non-regular hexagon), inherits the invariance. Regularity is convenient but not required; what matters is the angular arrangement of the gliders.

8 Design guidance for a six-glider array

With six gliders one could deploy a single regular hexagon or, say, two triangles. Section 4 appears to tie a single triangle with the hexagon, but §5 shows that tie to be an artifact of a symmetric placement; averaged over placement the triangle is the worst shape tested. Four arguments point the same way:

1. **Exposure at the lowest wavenumber.** In general flow the triangle is exposed at $m = 3$, lower than any other shape here, and a centered feature—a jet or eddy core the array straddles—drives exactly that mode (Fig. 4). The hexagon rejects everything below $m = 6$.

array	n	blind modes	centered placement (§4) lowest active m	centered placement (§4) swing ($\times 10^{-7}\ \mathrm{s}^{-1}$)	general placement (§5) lowest active m	general placement (§5) mean swing ($\times 10^{-7}\ \mathrm{s}^{-1}$)
triangle	3	$3, 6, 9, \ldots$	6	3.66	3	8.64
square	4	$4, 8, 12, \ldots$	4	9.81	4	6.12
pentagon	5	$5, 10, 15, \ldots$	10	0.15	5	4.04
hexagon	6	$6, 12, 18, \ldots$	6	3.66	6	3.09
octagon	8	$8, 16, 24, \ldots$	8	3.02	8	1.60

Table 2: Why the ranking changes between §4 and §5. At the centered placement the flow reverses across the array, so only even wavenumbers are present and an odd-n array is not exposed until $m = 2n$; the measured swing then orders by lowest active mode ($m = 4$ loudest, $m = 10$ quietest), not by n. At a general placement odd wavenumbers are present too, every array is exposed at $m = n$, and the swing orders by glider count.

2. **Noise.** The hexagon has half the variance of a triangle's divergence estimate (§7) and, being over-determined, is not at the mercy of a single bad sample.
3. **Robustness.** Losing one glider leaves the hexagon with five points (a sound plane fit); it leaves a triangle with two (no fit at all).
4. **Footprint and bias.** The hexagon spans about twice the area of an inscribed triangle and, with six points, can fit curvature to attack the uniform-mode bias that three points cannot resolve.

Two equilateral triangles at the same radius rotated by 60° *are* a regular hexagon; at different radii they leave a three-lobed blind spot and are strictly worse. The recommended six-glider footprint is the regular hexagon—the shape already adopted by the OSSE's symmetric-array family.

9 Summary

A glider array's plane-fit divergence is an azimuthal sampler. Worked from the specific to the general: a single-scale flow fools exactly one array shape and leaves the rest exactly invariant; a broadband flow at one symmetric placement produces a ranking that is not monotone in glider number, with a pentagon beating a hexagon and a triangle tying it; and averaging over where the array sits in the flow—the case that matters for a real deployment—restores a strictly monotone ordering, more gliders being better, with the triangle worst. The mechanism behind all three is that n equally spaced gliders cannot resolve structure repeating a multiple of n times around the ring, so the array is blind at $m = n, 2n, \ldots$ and its orientation error is set by the flow's energy at the lowest of these. Whether the velocity reverses across the array center decides whether the flow carries even or odd wavenumbers, which is what makes the symmetric placement special and its ranking misleading. Noise, robustness, and footprint all favor spreading six gliders into a regular hexagon.

Appendix: the aliasing condition

For completeness: writing the plane-fit divergence as a weighted sum of the sampled velocities and expanding the flow in azimuthal modes $e^{\mathrm{i}m\varphi}$, the estimate picks up the flow's wavenumber-m amplitude weighted by $\sum_k e^{\mathrm{i}m\psi_k}$, the sum over glider angles ψ_k. For n gliders at equal spacing this sum vanishes unless m is a multiple of n, which is the statement of §6 and Eq. (3). (Equivalently, the array's orientation error is governed by the vertex sums $\sum_k (x_k + \mathrm{i}y_k)^m$, which vanish for a regular n-gon unless n divides m.)